LEVEL 12 · PART VI

Advanced Networking

12

Goal: Understand the network layers, protocols and traffic-management patterns behind large-scale production systems.

RECOMMENDED TIME
3-4 weeks

TOPICS
14

PRACTICE / CAPSTONE
Build a lab: Envoy/Nginx with TLS termination, mTLS between two services, rate limiting and a circuit breaker, and observe it with tcpdump/curl.

Topics

#	Topic	What you will learn
1	OSI and TCP/IP Models **p.2323**	layers, encapsulation, where each tool/product lives
2	BGP Fundamentals **p.2335**	autonomous systems, route advertisement, anycast, famous BGP outages
3	DNS Architecture **p.2349**	anycast resolvers, split-horizon, GeoDNS, DNSSEC, DNS failures at scale
4	TLS **p.2365**	PKI, cert chains, ACME/Let's Encrypt, cipher suites, rotation
5	HTTP/2, HTTP/3 and QUIC **p.2381**	multiplexing, head-of-line blocking, QUIC over UDP, 0-RTT
6	Load Balancing **p.2399**	round-robin, least-conn, consistent hashing, health checks, global LB
7	CDNs **p.2415**	edge caching, cache keys, purge, origin shield, Cache-Control
8	VPNs **p.2431**	IPsec vs WireGuard, site-to-site, client VPN, bastion vs VPN vs zero-trust
9	Service Mesh **p.2445**	sidecar vs ambient, control/data plane, when it is worth it
10	Connection Pooling **p.2463**	keep-alive, pool sizing, PgBouncer, pool exhaustion
11	TLS Termination **p.2479**	edge vs passthrough vs re-encrypt, offload trade-offs

#	Topic	What you will learn
12	mTLS **p.2497**	mutual auth, client certs, SPIFFE/SPIRE, cert rotation
13	Rate Limiting **p.2513**	token bucket, leaky bucket, sliding window, distributed limiting with Redis
14	Circuit Breakers **p.2533**	closed/open/half-open, retries with backoff+jitter, bulkheads, timeouts
★	Level capstone project **p.2553**	Build a lab: Envoy/Nginx with TLS termination, mTLS between two services, rate limiting and a circuit breaker, and observe it with tcpdump/curl.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

